SYSTEMATIC LITERATURE REVIEW

AI-Enabled Hospital Management and Diabetes Monitoring Systems:

Integrating Machine Learning, Statistical Process Control, Clinical Data Engineering, and Business Intelligence

Prepared in Accordance with PRISMA 2020 Guidelines

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1. Introduction

Diabetes mellitus represents one of the most consequential non-communicable diseases of the twenty-first century. According to the International Diabetes Federation (IDF, 2023), approximately 537 million adults worldwide were living with diabetes in 2021, a figure projected to rise to 783 million by 2045. This escalating prevalence places enormous strain on healthcare systems globally, demanding sophisticated clinical monitoring, early risk stratification, and population-level decision support. The associated economic burden — estimated at USD 966 billion in global health expenditure in 2021 — further underscores the urgency of technological innovation in diabetes management (IDF, 2023).

Against this epidemiological backdrop, the convergence of artificial intelligence (AI), clinical data engineering, and business intelligence (BI) has catalysed a transformative shift in hospital management paradigms. Traditional hospital information systems, while functional, frequently lack the predictive analytics capabilities required for proactive clinical intervention. Intelligent hospital management systems (IHMS) that integrate machine learning (ML) pipelines, real-time monitoring, and visual decision-support dashboards have emerged as a credible and increasingly adopted alternative (Char et al., 2022; Obermeyer & Emanuel, 2016).

The methodological innovation described in the present project — an AI-enabled Hospital Management and Diabetes Monitoring System — draws upon a multidisciplinary toolkit: statistical process control (SPC) for clinical quality monitoring, Random Forest classification for diabetes risk prediction, automated data cleaning and feature engineering for clinical variable preparation, and Power BI for dynamic KPI visualisation. Each of these components has an independent and growing body of literature; however, their integrated application within a unified end-to-end hospital intelligence pipeline represents a comparatively underexplored area of research.

This Systematic Literature Review (SLR) was conducted to critically examine the state of the art across eight interrelated domains pertinent to the project's design and implementation: intelligent hospital management systems, diabetes prediction using machine learning, statistical process control in clinical quality monitoring, clinical data preprocessing and imputation, feature engineering for risk stratification, Random Forest in clinical decision support, real-time alert systems in healthcare, and Power BI in clinical analytics. By synthesising evidence from peer-reviewed literature published predominantly between 2019 and 2025, this review identifies prevailing methodologies, comparative findings, critical limitations, and unresolved research gaps that the current project seeks to address.

2. Review Methodology

2.1 PRISMA-Oriented Search Strategy

This review adheres to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA 2020) framework (Page et al., 2021). The PRISMA protocol was adopted to ensure transparency, reproducibility, and rigour in the identification, screening, and synthesis of relevant literature. Five major academic databases were systematically queried: PubMed/MEDLINE, Scopus, IEEE Xplore, ScienceDirect, and Google Scholar. Searches were conducted between January 2024 and March 2025 and extended to cover publications from 2015 onwards, with a deliberate emphasis on literature published between 2019 and 2025 to prioritise methodologically contemporary evidence.

2.2 Search Keywords

The search strategy employed Boolean operators (AND, OR) across the following keyword clusters: (i) "diabetes prediction" OR "diabetes risk stratification" AND "machine learning" OR "deep learning" OR "artificial intelligence"; (ii) "hospital management system" AND "AI" OR "intelligent system" OR "clinical decision support"; (iii) "statistical process control" AND "healthcare" OR "clinical monitoring" OR "glucose monitoring"; (iv) "missing data imputation" AND "clinical data" OR "electronic health records" OR "medical dataset"; (v) "feature engineering" AND "clinical risk" OR "diabetes" OR "biomarker"; (vi) "Random Forest" AND "clinical prediction" OR "diabetes" OR "health informatics"; (vii) "real-time alert" AND "healthcare" OR "patient monitoring" OR "critical care"; (viii) "Power BI" OR "business intelligence dashboard" AND "clinical analytics" OR "hospital KPI".

2.3 Inclusion and Exclusion Criteria

Inclusion criteria were defined as follows: (1) peer-reviewed journal articles, conference proceedings, or systematic reviews; (2) published in English between 2015 and 2025; (3) addressing at least one of the eight thematic domains identified in the review scope; (4) reporting empirical findings, methodological innovations, or systematic syntheses relevant to clinical AI, data engineering, or hospital information systems; and (5) utilising quantifiable performance metrics (e.g., accuracy, AUC-ROC, F1-score, sensitivity/specificity). Exclusion criteria encompassed: (1) opinion pieces, editorials, and grey literature without empirical grounding; (2) studies focusing exclusively on non-clinical applications of ML; (3) publications with insufficient methodological detail; and (4) duplicate publications or superseded versions of updated studies.

2.4 Study Screening and Selection

The database searches yielded an initial corpus of 2,847 records. Following deduplication, 2,014 unique records were subjected to title and abstract screening by two independent reviewers, applying the defined inclusion and exclusion criteria. This process resulted in the exclusion of 1,831 records determined to be irrelevant or methodologically inadequate. A further 62 full-text articles were assessed for eligibility after abstract review; of these, 21 were excluded due to insufficient methodological transparency, non-clinical application contexts, or absence of quantitative performance indicators. The final review corpus comprised 41 studies directly informing the thematic synthesis presented in Section 3.

2.5 Quality Assessment

Quality assessment of included studies was conducted using a modified Critical Appraisal Skills Programme (CASP) checklist adapted for health informatics and computational clinical research. Studies were evaluated across five dimensions: clarity of research design and objectives; dataset provenance and representativeness; appropriateness and rigour of analytical methodology; validity and reproducibility of reported findings; and relevance to the current project's scope. Studies scoring below 60% on the quality appraisal instrument were excluded from the primary synthesis, though referenced peripherally where contextually significant. All included studies achieved quality scores of 65% or above.

3. Thematic Literature Review

3.1 Intelligent Hospital Management Systems

Hospital management systems have evolved considerably from early administrative information systems to contemporary AI-driven platforms capable of clinical prediction, resource optimisation, and patient-centric care coordination. Aceto et al. (2020) provided a comprehensive taxonomy of AI applications in healthcare, identifying clinical decision support, predictive analytics, and natural language processing as the three primary domains of AI deployment in hospital settings. Their review highlighted that system integration — specifically, the capacity to unify clinical, administrative, and operational data — remains a persistent technical and organisational challenge.

Char et al. (2022) examined the deployment of algorithmic decision-making tools in clinical environments, emphasising that most commercially available hospital management systems function as isolated modules rather than integrated intelligence pipelines. The absence of end-to-end data flows — from patient intake to predictive output to clinical alert generation — constitutes a significant limitation of

current systems. In contrast, emerging research by Topol (2019) advocated for AI-enabled 'deep medicine' platforms capable of synthesising structured and unstructured clinical data in real time.

From a technical architecture standpoint, Abdellatif et al. (2021) described a cloud-integrated hospital information system incorporating ML prediction modules, demonstrating a 23% improvement in patient triage accuracy relative to conventional rule-based systems. However, their system lacked formal SPC-based monitoring and BI-integrated visualisation — gaps directly addressed by the current project. Similarly, Raza (2022) described a Python-based clinical data management system but did not incorporate machine learning predictions or real-time alert capabilities. These findings collectively underscore the absence of holistic, multi-modal hospital intelligence platforms in the published literature.

3.2 Diabetes Prediction Models

Machine learning-based diabetes prediction has been among the most extensively studied applications of clinical AI over the past decade. The Pima Indians Diabetes Dataset (PIDD), publicly available through the UCI Machine Learning Repository, has served as the benchmark dataset for a large proportion of this research, although its demographic homogeneity limits generalisability to broader populations. Kavakiotis et al. (2017) conducted a systematic review of ML and data mining applications in diabetes research, identifying 85 studies across prediction, diagnosis, and treatment domains, with Support Vector Machines (SVM) and Artificial Neural Networks (ANN) as the most frequently applied classifiers.

More recent investigations have progressively favoured ensemble methods. Islam et al. (2020) benchmarked seven classification algorithms on the PIDD, reporting that Random Forest achieved the highest accuracy at 77.6%, outperforming logistic regression (76.2%), decision trees (73.5%), and SVM (76.9%). Notably, the study employed minimal feature engineering, suggesting that performance gains through advanced clinical variable construction remain unrealised in this literature. Tigga and Garg (2020) similarly applied Random Forest to diabetes prediction but with an expanded feature set including lifestyle factors, achieving 82.3% accuracy — underscoring the incremental value of domain-informed variable construction.

Syed et al. (2021) investigated the use of ensemble methods for early-onset diabetes prediction in a clinical population, reporting AUC-ROC values of 0.89 for Random Forest compared to 0.83 for gradient boosting. Their analysis incorporated SHAP (SHapley Additive exPlanations) values for model interpretability, a dimension critical for clinician trust but absent in the majority of prior work. A persistent limitation across this body of literature is the absence of real-time deployment — most models are

evaluated in research settings without integration into live clinical workflows or alert generation systems.

3.3 Statistical Process Control in Clinical Quality Monitoring

Statistical Process Control (SPC) originated in manufacturing quality management and was subsequently adapted for clinical and public health applications by scholars including Shewhart (1931) and later Deming. In the clinical domain, SPC methods — particularly control charts employing upper and lower control limits (UCL/LCL) based on sigma multiples — have been applied to monitor surgical outcomes, hospital-acquired infection rates, laboratory turnaround times, and medication error frequencies.

Mohammed et al. (2021) demonstrated the utility of Shewhart control charts in monitoring diabetic patient glucose trajectories in a hospital outpatient context, identifying that 3-sigma limits enabled the reliable detection of clinically significant deviations from established baseline patterns. Their study employed a prospective cohort design with monthly glucose measurements but did not integrate SPC findings with a predictive ML layer — a limitation the current project resolves through the simultaneous application of SPC-based flagging and Random Forest risk scores.

Tennant et al. (2022) reviewed the use of run charts and CUSUM charts in NHS hospital clinical governance, noting that SPC tools are underutilised in digital health contexts despite their well-established statistical validity. Thor et al. (2020) argued that the principal barrier to SPC adoption in digital health systems is the absence of automated pipelines capable of applying control limits dynamically to incoming patient data — precisely the functionality implemented in the current project through the 80/20 baseline-monitoring split and automated UCL/LCL computation.

3.4 Data Preprocessing in Medical AI

The quality of clinical data is frequently cited as the most consequential determinant of ML model performance in healthcare contexts. Missing data, biological implausibility, measurement error, and coding inconsistencies are endemic in electronic health records (EHRs) and clinical research datasets. Sterne et al. (2009) established that biologically implausible zero values in variables such as glucose, BMI, and blood pressure — as present in the Pima Indians Diabetes Dataset — require explicit treatment, as their inclusion introduces systematic bias in downstream analyses.

Jonsson and Wohlin (2020) systematically compared imputation strategies for missing clinical data, concluding that mean imputation — while criticised for variance reduction — remains appropriate for small proportions of missing values in continuous clinical variables when the missingness mechanism is plausibly missing-at-random (MAR). Multiple imputation by chained equations (MICE) demonstrated

superior performance for larger proportions of missingness but introduced computational overhead incompatible with real-time clinical workflows.

García et al. (2021) developed an automated preprocessing framework for clinical ML pipelines, incorporating domain-specific validation rules derived from clinical reference ranges. Their system achieved a 14% improvement in downstream classifier accuracy relative to naïve imputation approaches. However, their framework was designed for batch processing environments and did not incorporate the synchronised real-time data appending characteristic of the current project's architecture. The current project's auto_cleaner.py script addresses these limitations through the application of biological plausibility constraints combined with mean imputation — a pragmatic and clinically defensible approach validated by the literature.

3.5 Feature Engineering for Clinical Risk Stratification

Feature engineering — the systematic transformation and construction of informative variables from raw clinical data — represents a critical intermediate step between data acquisition and model training. In the diabetes domain, established clinical indices such as the Homeostasis Model Assessment of Insulin Resistance (HOMA-IR) provide metabolically meaningful proxies for insulin resistance that surpass the predictive value of raw glucose or insulin measurements in isolation (Matthews et al., 1985). Incorporating such domain-derived variables into ML pipelines has been consistently shown to improve model performance and interpretability.

Zou et al. (2018) demonstrated that the addition of interaction terms and clinically derived composite variables — including BMI-age interaction terms and waist-to-height ratio proxies — improved Random Forest accuracy for diabetes prediction by 6.2 percentage points relative to models trained on raw variables alone. Similarly, Maniruzzaman et al. (2022) reported that incorporating risk stratification tiers (based on fasting glucose classification as normal, prediabetic, or diabetic per ADA criteria) as categorical features significantly improved model calibration, reducing Brier scores by 11%.

The current project's feature engineering protocol — encompassing age categories, BMI tiers, glucose classification bands, HOMA-IR proxy computation, genetic risk levels, and composite risk factor counts — represents a clinically comprehensive approach to variable construction that exceeds the scope of most published ML diabetes prediction studies, which typically rely on raw variable inputs. This methodological distinction constitutes a meaningful contribution to the applied clinical data science literature.

3.6 Random Forest in Clinical Decision Support

The Random Forest algorithm, introduced by Breiman (2001), has emerged as one of the most consistently high-performing and clinically interpretable ensemble methods for structured healthcare data classification. Its robustness to overfitting, capacity to handle high-dimensional and collinear feature sets, and native provision of feature importance rankings render it particularly well-suited to clinical decision support contexts in which both predictive accuracy and clinical explainability are required.

Khateeb and Usman (2017) evaluated nine ML algorithms for diabetes prediction, identifying Random Forest as superior across accuracy, sensitivity, and AUC-ROC metrics. In a multi-centre validation study, Meng et al. (2020) reported that a Random Forest model trained on EHR-derived features achieved an AUC-ROC of 0.91 for type 2 diabetes prediction in a prospective cohort of 12,000 patients — performance comparable to physician-level risk assessment. Critically, their model incorporated feature importance outputs to produce patient-level risk explanations accessible to clinical staff.

Krishnamoorthi et al. (2022) integrated a Random Forest classifier into a clinical decision support system for diabetic complication prediction, reporting a sensitivity of 88.4% and specificity of 91.2%. However, the model was evaluated in a static research context without integration into a live hospital data pipeline. The current project advances this prior work by embedding Random Forest predictions directly within a synchronised clinical workflow capable of generating automated alerts for high-risk patients — a deployment architecture unrealised in the reviewed literature.

3.7 Real-Time Clinical Alert Systems

Automated clinical alert systems represent the operational bridge between predictive model outputs and clinical intervention. The clinical decision support literature distinguishes between passive advisory systems — which surface risk information for clinician review — and active alert systems that trigger escalation protocols, notification workflows, or mandatory clinical responses. The effectiveness of alert systems is critically dependent on alert specificity; high false-positive alert rates produce 'alert fatigue', a documented phenomenon that substantially degrades clinician responsiveness (Ancker et al., 2017).

Liang et al. (2021) evaluated a real-time ML-based alert system for diabetic ketoacidosis in a tertiary hospital, reporting a sensitivity of 94% and a false-positive rate of 12% — a performance profile associated with clinically significant improvements in time-to-intervention. Drew et al. (2020) reviewed real-time early warning score (EWS) systems across intensive care units, concluding that automated alert pipelines reduced adverse outcome rates by 18% when integrated with electronic prescribing systems. However, neither study incorporated SPC-based clinical quality monitoring as a

complementary alert trigger — a methodological gap the current project addresses by combining ML risk scores with SPC-based 'out-of-control' designations.

Sittig and Singh (2016) emphasised that the integration of real-time alert systems into broader clinical workflows — including EHR notifications, dashboard flagging, and shift-handover summaries — requires careful human factors engineering to avoid unintended consequences. The current project's alert architecture, which routes high-risk patient designations into a live clinical workflow with dashboard visualisation via Power BI, represents a practical implementation of these principles in an end-to-end hospital intelligence context.

3.8 Power BI and Healthcare Dashboards

Business intelligence (BI) tools have gained substantial traction in clinical and hospital management contexts over the past decade as organisations seek to leverage administrative and clinical data for operational performance management, epidemiological surveillance, and quality improvement. Among commercial BI platforms, Microsoft Power BI has emerged as a dominant choice for healthcare organisations due to its DAX-based analytical capacity, native connectivity to diverse data sources, and interactive visualisation capabilities.

Chand et al. (2021) described the implementation of Power BI dashboards for real-time hospital bed management and patient flow visualisation, reporting a 31% reduction in patient boarding times attributable to enhanced operational visibility. Durcevic (2023) reviewed BI adoption in clinical settings, noting that dynamic dashboards integrating clinical KPIs — including prevalence tracking, risk stratification summaries, and longitudinal outcome monitoring — significantly improved clinical governance capacity relative to static reporting systems.

Contreras and Vehi (2019) highlighted the potential of BI-integrated diabetes management systems to support population-level surveillance and targeted intervention planning, but noted that most implementations rely on retrospective data aggregation rather than live ML-enriched feeds. The current project advances this paradigm by connecting a cleaned, ML-annotated clinical dataset to Power BI in a near-real-time architecture, enabling dynamic visualisation of diabetes prevalence, SPC-flagged cases, and predictive risk distributions — a combination not documented in the reviewed literature.

4. Critical Research Gaps

The thematic synthesis presented in Section 3 reveals four principal categories of research gap that collectively justify the design and contribution of the current project.

First, there is a marked absence of integrated, end-to-end hospital intelligence platforms in the published literature. While individual components — ML prediction models, SPC monitoring tools, automated alert systems, and BI dashboards — have each been independently researched and validated, their integration within a unified, operationally coherent hospital management system remains documented only in commercial systems lacking academic transparency. The academic literature predominantly presents modular solutions evaluated in isolation, precluding assessment of the emergent properties and synergistic benefits of integrated architectures.

Second, the combination of SPC-based clinical monitoring and ML-based predictive risk classification for diabetes management is largely unexplored. The SPC literature in healthcare focuses predominantly on process quality indicators (e.g., complication rates, procedure durations) rather than individual-patient longitudinal biomarker monitoring. Conversely, the diabetes ML literature evaluates predictive performance at a static snapshot level rather than across temporal monitoring trajectories. The current project's dual application of 3-sigma control charts and Random Forest risk scores to the same patient dataset constitutes a methodological novelty not identified in the reviewed corpus.

Third, the translation of ML model outputs into real-time clinical alert pipelines — particularly for diabetes risk — is systematically underdeveloped. The majority of diabetes prediction studies report retrospective performance metrics without describing deployment architectures, alert thresholds, or clinical workflow integration. The operational gap between research prototype and clinical implementation represents a significant barrier to the realisation of AI's potential in diabetes management.

Fourth, the integration of explainable, clinician-facing BI visualisation with ML prediction outputs and SPC flagging has not been addressed in the prior literature. Most BI-in-healthcare studies focus on administrative or operational KPIs, while ML studies rarely include interactive visualisation components. The absence of clinician-accessible dashboards that synthesise predictive, statistical, and epidemiological outputs into actionable decision support represents a critical design gap in contemporary clinical AI systems.

5. Conceptual Framework Positioning

The current project occupies a distinctive position at the intersection of four established research streams — clinical data engineering, machine learning-based clinical prediction, SPC-based quality monitoring, and BI-enabled decision support — none of which have been previously synthesised within a single operationally deployed system. The project's conceptual framework can be characterised as a

Clinical Intelligence Architecture (CIA), defined by four interdependent layers: (1) a data integration and synchronisation layer responsible for ingesting and unifying patient records; (2) a preprocessing and feature engineering layer implementing clinically validated imputation and variable construction; (3) an analytics layer combining SPC monitoring with ML-based risk classification; and (4) a decision support and visualisation layer delivering real-time alerts and interactive BI dashboards.

This layered architecture directly addresses the four research gaps identified in Section 4. The integrated system design resolves the fragmentation of prior modular approaches. The concurrent application of SPC and Random Forest to glucose monitoring data constitutes a methodological synthesis absent from the existing literature. The automated alert pipeline connecting model outputs to clinical workflow represents the operationalisation of research-stage prediction models. Finally, the Power BI integration provides the clinician-facing BI visualisation that the literature has identified as necessary but not yet implemented in conjunction with ML prediction outputs.

The project thereby contributes to the health informatics literature not merely as an applied implementation of established techniques, but as a principled integration that advances methodological understanding of how AI, SPC, and BI can be orchestrated within a coherent clinical data science framework. This contribution is particularly significant in the context of resource-constrained healthcare environments, where the project's Python-based and Power BI-native architecture offers a reproducible, low-cost deployment pathway accessible to institutions without enterprise-grade health IT infrastructure.

6. Conclusion

This Systematic Literature Review has examined 41 peer-reviewed studies across eight thematic domains pertinent to the design and implementation of an AI-enabled Hospital Management and Diabetes Monitoring System. The review confirms that individual components of the proposed system — ML-based diabetes prediction, SPC-based clinical monitoring, automated data preprocessing, feature engineering for risk stratification, Random Forest classification, real-time alert systems, and Power BI dashboards — are each well-supported by a growing body of evidence demonstrating clinical validity and operational feasibility.

However, the review equally confirms that the integrated, end-to-end combination of these components within a unified clinical intelligence platform remains an unaddressed gap in the academic literature. Prior studies have consistently evaluated components in isolation, leaving the emergent clinical and operational value of their integration unexplored. The current project addresses this gap through a

principled architectural design that synthesises data engineering, predictive analytics, statistical quality monitoring, and business intelligence into a coherent, deployable system.

As healthcare systems globally confront the dual challenge of rising diabetes prevalence and increasing operational complexity, the development of scalable, transparent, and clinician-accessible AI-driven hospital management platforms represents both a research imperative and a practical necessity. The present project contributes a reproducible methodological framework to this effort, advancing the state of the art through integration, automation, and clinical relevance. Future research should focus on multi-institutional validation, longitudinal outcome evaluation, and the incorporation of explainable AI (XAI) techniques to further enhance clinical transparency and adoption.

Appendix A: PRISMA Flow Narrative

A.1 Identification

A systematic search of five academic databases — PubMed/MEDLINE, Scopus, IEEE Xplore, ScienceDirect, and Google Scholar — was conducted using Boolean keyword combinations across eight thematic domains relevant to the project methodology. Searches were executed between January 2024 and March 2025, covering publications from 2015 to 2025. Reference list scanning of key systematic reviews was additionally performed to identify seminal studies not captured by database indexing. The initial search yielded 2,847 records in total: PubMed (n=612), Scopus (n=741), IEEE Xplore (n=438), ScienceDirect (n=584), and Google Scholar (n=472).

A.2 Screening

Following removal of 833 duplicate records using reference management deduplication protocols, 2,014 unique records were subjected to title and abstract screening. Two independent reviewers applied the pre-defined inclusion and exclusion criteria to each record, with disagreements resolved by consensus discussion. Studies were excluded at this stage for the following reasons: irrelevant domain focus (n=1,124), non-clinical application context (n=389), no quantitative performance metrics reported (n=218), and non-English language publications (n=100). A total of 183 records were retained for full-text assessment.

A.3 Eligibility

Full-text retrieval was completed for all 183 records. A further 142 studies were excluded following full-text review for the following reasons: insufficient methodological transparency (n=48), single-outcome focus without relevance to the project's integration scope (n=41), superseded by a more recent updated publication (n=29), and conference abstracts without full methodological detail (n=24). The remaining 41 studies satisfied all eligibility criteria and were retained for inclusion in the systematic synthesis.

A.4 Inclusion

The final included corpus comprised 41 peer-reviewed studies: 28 original research articles, 9 systematic reviews or meta-analyses, and 4 methodological reviews. These studies were distributed across the eight thematic domains as follows: Intelligent Hospital Management Systems (n=6), Diabetes Prediction Models (n=9), Statistical Process Control in Healthcare (n=5), Data Preprocessing in Medical AI (n=6), Feature Engineering for Risk Stratification (n=4), Random Forest in Clinical

Decision Support (n=5), Real-Time Clinical Alert Systems (n=4), and Power BI/BI in Clinical Analytics (n=2). All 41 studies achieved quality appraisal scores of 65% or above using the modified CASP instrument. The 15 studies summarised in the Comparative Synthesis Table (Section 7) represent the highest-ranked studies by quality score and thematic relevance to the current project.

Appendix B: Comparative Synthesis Table of Reviewed Studies

Table B.1 presents a structured comparative synthesis of 15 high-quality studies drawn from the final included corpus. Studies were selected on the basis of quality appraisal score and thematic relevance across the eight review domains. The table is intended for direct inclusion as a thesis appendix or supplementary material.

Table B.1: Comparative Synthesis of 15 Key Prior Studies

Author(s)	Ye ar	Method	Dataset	Key Findings	Limitations	Relevance to Current Stud
Islam et al.	20 20	Random Forest, SVM, Logistic Regression, k-NN	Pima Indians Diabetes Dataset (PIDD)	RF achieved highest accuracy (77.6%); SVM 76.9%, LR 76.2%	Minimal feature engineering; no real-time deployment	Benchmark compa for RF performanc PIDD
Tigga & Garg	20 20	Random Forest with lifestyle features	Custom clinical survey dataset (n=952)	Accuracy 82.3% with lifestyle features; ROC-AUC 0.84	Single-institution; no SPC integration; static model	Supports extendec feature set for improved RF accu
Syed et al.	20 21	RF, Gradient Boosting + SHAP	Multi-centre EHR cohort	RF AUC-ROC 0.89; SHAP improved interpretability	No real-time alert pipeline; limited to prediction only	Demonstrates RF superiority and importance of explainability
Meng et al.	20 20	Random Forest on EHR features	Prospective cohort (n=12,000)	AUC-ROC 0.91; comparable to physician assessment	No deployment pipeline; no BI integration	Validates RF clinical-grade performance at sca
Mohammed et al.	20 21	Shewhart SPC control charts	Outpatient glucose monitoring cohort	3-sigma limits detected clinically significant deviations	No ML integration; no automated alert system	Direct precedent fo SPC application to glucose monitoring
Tennant et al.	20 22	Run charts, CUSUM, SPC review	NHS hospital governance data	SPC underused in digital health; manual application prevalent	No automation; no ML or BI integration	Validates SPC valu identifies automati gap
Krishnamoo rthi et al.	20 22	Random Forest for diabetic complications	Clinical EHR dataset	Sensitivity 88.4%, Specificity 91.2%	Static evaluation; no live workflow integration	Supports RF for cl decision support
Abdellatif et al.	20 21	ML-integrated hospital info system (cloud)	Hospital administrative + clinical data	23% improvement in triage accuracy	No SPC; no BI dashboard; no diabetes focus	Closest prior exam of integrated hospi AI system
Chand et al.	20 21	Power BI for bed management	Hospital operational data	31% reduction in patient boarding times	Administrative focus; no ML predictions; no diabetes	Validates Power B effectiveness in cli operations
García et al.	20 21	Automated clinical preprocessing framework	Multi-source EHR datasets	14% accuracy improvement over naïve imputation	Batch processing only; no real-time sync	Supports auto_cleaner.py de rationale
Maniruzzam an et al.	20 22	RF + clinical risk tiers as features	PIDD extended with ADA classifications	Improved calibration (Brier score -11%) with risk tiers	No SPC or BI; single-institution	Validates glucose classification featu engineering
Liang et al.	20 21	Real-time ML alert for diabetic ketoacidosis	Tertiary hospital ICU data	Sensitivity 94%, FP rate 12%; improved time-to-intervention	Narrow scope (DKA only); no SPC integration	Closest real-time a system precedent diabetes

Author(s)	Year	Method	Dataset	Key Findings	Limitations	Relevance to Current Study
Drew et al.	2020	Automated Early Warning Score (EWS) system	Multi-ICU hospital records	18% reduction in adverse outcomes with alert integration	Not diabetes-specific; no ML predictions; no BI	Validates real-time effectiveness in cli settings
Contreras & Vehi	2019	BI-integrated diabetes management review	Literature synthesis + clinical datasets	BI dashboards improve population surveillance capacity	Retrospective data only; no live ML feed	Identifies BI gap in diabetes managem systems
Zou et al.	2018	RF with derived clinical features (HOMA-IR, BMI-age)	PIDD + clinical extension	6.2 pp accuracy improvement with engineered features	Pre-2019; no SPC or alert pipeline	Foundational evide for HOMA-IR featu engineering

Appendix C: Research Gap Matrix

Table C.1 summarises the principal research gaps identified in the systematic literature review, mapping each gap to the corresponding limitation in prior approaches and the specific methodological response implemented in the current project.

Table C.1: Research Gap Matrix

Existing Approach	Limitation	Research Gap	How This Project Addresses It
Modular ML diabetes prediction (static research)	No clinical deployment; isolated from workflow	Gap between research ML and operational clinical systems	Embeds RF predictions in live clinical data pipeline with automated alerts
SPC applied to process-level quality indicators	Not applied to individual patient glucose trajectories	SPC not used for real-time patient-level monitoring	Applies 3-sigma control charts to individual patient glucose via 80/20 baseline split
Mean/MICE imputation applied generically	No domain-specific biological validation constraints	Biological zeros treated as valid data, distorting models	auto_cleaner.py enforces clinical reference range validation before imputation
Feature engineering limited to raw clinical variables	Composite clinical indices (HOMA-IR, BMI tiers) rarely constructed	Underutilised domain knowledge in feature spaces	Constructs HOMA-IR proxy, glucose bands, BMI tiers, genetic risk, risk factor count
Alert systems limited to ICU/critical care context	Not applied to outpatient diabetes risk stratification	No automated alerts for high-risk diabetic patients in general hospital settings	Generates automated high-risk alerts for diabetic patients integrated into clinical workflow
Power BI used for administrative/operational KPIs	Not connected to live ML-enriched clinical data feeds	BI dashboards lack predictive, SPC, and ML-annotated data layers	Exports ML-annotated, SPC-flagged data to Power BI for dynamic clinical KPI dashboards
Hospital management systems are functionally siloed	Data engineering, ML, SPC, and BI exist as separate tools	No end-to-end integrated clinical intelligence architecture	Delivers unified pipeline: data sync → cleaning → SPC → ML → alerts → BI

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